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A comparison between centralized and asynchronous federated learning approaches for survival outcome prediction using clinical and PET data from non-small cell lung cancer patients

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ABSTRACT

Background and Objective: Survival analysis plays an essential role in the medical field for optimal treatment decision-making. Recently, survival analysis based on the deep learning (DL) approach has been proposed and is demonstrating promising results. However, developing an ideal prediction model requires integrating large datasets across multiple institutions, which poses challenges concerning medical data privacy.
Methods: In this paper, we propose FedSurv, an asynchronous federated learning (FL) framework designed to predict survival time using clinical information and positron emission tomography (PET)-based features. This study used two datasets: a public radiogenic dataset of non-small cell lung cancer (NSCLC) from the Cancer Imaging Archive (RNSCLC), and an in-house dataset from the Chonnam National University Hwasun Hospital (CNUHH) in South Korea, consisting of clinical risk factors and F-18 fluorodeoxyglucose (FDG) PET images in NSCLC patients. Initially, each dataset was divided into multiple clients according to histological attributes, and each client was trained using the proposed DL model to predict individual survival time. The FL framework collected weights and parameters from the clients, which were then incorporated into the global model. Finally, the global model aggregated all weights and parameters and redistributed the updated model weights to each client. We evaluated different frameworks including single-client-based approach, centralized learning and FL.
Results: We evaluated our method on two independent datasets. First, on the RNSCLC dataset, the mean absolute error (MAE) was 490.80 ± 22.95 d and the C-Index was 0.69 ± 0.01. Second, on the CNUHH dataset, the MAE was 494.25 ± 40.16 d and the C-Index was 0.71 ± 0.01. The FL approach achieved centralized method performance in PET-based survival time prediction and outperformed single-client-based approaches.
Conclusions: Our results demonstrated the feasibility and effectiveness of employing FL for individual survival prediction in NSCLC patients, using clinical information and PET-based features.

1. Introduction

Deep learning (DL) [1] approaches have been actively investigated in various medical fields, such as pathology, genomics, and radiology analysis [2]. The successful development of a high-performance DL prediction model often requires the integration of a substantial amount of diverse data, encompassing clinical information, medical images, and genomic data, enabling comprehensive analysis and accurate. Moreover, to develop a generalized artificial intelligence (AI) model that can be applied to different patients from diverse medical institutions, various datasets from diverse hospitals must be utilized. However, acquiring medical data is difficult due to its high sensitivity and tightly regulated usage in terms of privacy. Although cooperation between multiple medical centers or institutions is essential, it is difficult to share, centralize, and integrate data between hospitals due to regulatory and legal concerns including privacy issues. To overcome these obstacles, a federated learning (FL) technique was used to train global and general AI models [3–5].

FL, a new research direction for AI, can be used to build personalized models without violating user-privacy [6]. With the FL applications, each client site uses local data for training and then uploads the model specifics (e.g., weights and gradients) to the global server for aggrega-

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tion, after which the server transfers the model or weight update back to the separate local site to achieve the desired learning goal. Moreover, FL is divided into three categories: horizontal FL, vertical FL, and federated transfer learning [6,7].

The potential of FL methods in healthcare has been demonstrated through recent successful applications in various medical domains [8,9], such as electronic health record analysis [10], magnetic resonance imaging [11–13] and ultrasound image analysis [14], electroencephalogram classification [15,16], disease detection based on electrocardiogram [17,18], and survival prediction [19,20].

Survival analysis is an important step in planning the correct cancer treatment. In this context, our previous studies [21,22] demonstrated the potential of using positron emission tomography (PET)-based convolutional neural network (CNN), such as DenseNet [23] and ResNet [24], in conjunction with the Cox proportional hazard model (CoxPH) [25], to predict individual survival distributions and times in non-small cell lung cancer (NSCLC) patients. PET is a molecular imaging technique that offers crucial insight into the molecular characteristics of disease through the use of radiopharmaceuticals. Specifically, F-18 fluorodeoxyglucose (FDG) PET plays a pivotal role in identifying metabolic alteration in cancer cells, which often exhibit increased glycolysis. FDG PET imaging is widely utilized in the initial staging [26], therapeutic monitoring [27,28]), and prognosis prediction [29–31] for cancer patients including NSCLC patients. Given that NSCLC constitutes the majority (85%) of lung cancer cases and is a leading cause of death worldwide [32], our approach of developing refined survival prediction models combining clinical risk factors with FDG PET images is instrumental in enhancing survival prediction models for NSCLC patients. Such models are crucial for improving patient outcomes and advancing global healthcare.

In this study, we proposed an asynchronous FL-based microservice to develop a high-performance and generalized survival prediction model for NSCLC patients without compromising privacy. Notably, the DL model was trained through asynchronous FL to achieve high performance close to the ideal benchmark with improved learning loss, learning speed, and robustness.

The main contributions are as follows:

- Demonstrated feasibility and effectiveness: We demonstrated the feasibility and effectiveness of FedSurv, an FL approach, for survival time prediction using a large-scale multimodal dataset. The proposed DeepFedSurv model is a DL model specifically designed for predicting survival time in cancer patients. By integrating FL with DL, we successfully addressed the challenge of survival time prediction for patients diagnosed with NSCLC.
- Demonstrated multimodal prediction: We demonstrated the effectiveness of a combination of PET-based features and clinical features for the multimodal prediction of survival time using two independent datasets. This highlights the potential benefits of utilizing both types of data in predicting patient outcomes.
- Addressed the demand for multimodal studies: We further explored the adoption of FL in multimodal domains, specifically combining PET and clinical data. This effort was undertaken to address the increasing demand for large-scale and inter-institutional studies, contributing to advancements in the field of medical research.

2. Related work

2.1. Concept of federated learning

According to the distribution characteristics of the data (sample space and feature space), FL can be divided into three categories: vertical federated learning, horizontal federated learning, and federated transfer learning [4].

Vertical FL or feature-based FL is used for cases in which each device contains the same sample instances but differs in feature space. In other words, vertical FL divides the datasets vertically (by feature dimension) and then takes out the part of data that has the same sample but features are not exactly the same for training. Vertical FL is to aggregate these different features in an encrypted state to enhance the ability of the model. At present, there are many machine learning (ML) models for vertical data partition such as classification [33], safe linear regression [34]. In medical applications, there are a few published vertical FL-based studies, such as an autoencoder-based ML model for vertically incomplete data [35].

Horizontal FL or sample-based FL is used for cases in which each device contains different sample instances but with the same feature space. Horizontal FL is to split the datasets horizontally (by the sample dimension) and then takes out the part of the data that features are the same, but samples are not exactly the same for training. In horizontal FL, it is common for all clients to calculate and upload local model weights so that the central server can aggregate them into a global model. In this respect, the horizontal FL framework can increase the total number of training samples and improve the accuracy of the model. Horizontal FL is quite common in FL applied for medical applications [36–40,14,41–44]. However, FL research in medical applications only handles a limit of 2 to 100 clients [45]. For instance, Li et al. [37] experimented with four medical institutions in different places for the autism spectrum disorder prediction scenario. Ming Y. Lu et al. [20] performed four sites for classifying primary Breast Invasive Carcinoma, four sites for multi-class classification of Renal Cell Carcinoma, and three sites for weakly-supervised survival prediction.

Federated transfer learning is applied to the scenarios in which the two clients differ in sample space and feature space. It is similar to transfer learning in ML, where new features are added to a pre-trained model. Federated Transfer Learning deals with problems exceeding the scope of existing federated learning algorithms. However, the research in federated transfer learning is still in the early stages, and there is a lot of room for improvement. One related study [46] proposed the FedHealth method to classify human activity and then transferred it to Parkinson's disease classification.

Here, we focus on horizontal federated learning, where the clients share the same feature space but different sample space.

2.2. Data distribution

Within an FL environment, data distribution is frequently classified into IID (Independent and Identically Distributed) and non-IID (non-Independent and Identically Distributed). Non-IID occurs commonly in the medical domain because of different medical tools manufacturers, calibrated techniques, and medical data acquisition techniques. Therefore, non-IID can result from an imbalance in data quantity, features, or labels. In the FL environment, the non-IID characteristics can be identified from four different forms such as (1) quantity distribution skew, (2) label distribution skew, (3) feature distribution skew, and (4) concept shift skew [47,8]. In the quantity skew characteristic, the class distribution of data instances is not equal or far from equal across clients in the FL scheme [37,36,48]. In the label distribution skew, the distribution of labels varies between different clients [3,49]. For the feature distribution skew characteristic, the distribution of features varies between different clients. Unlike the distribution of features in IID, which is the same across all clients, the distribution of features in non-IID varies between clients [50]. Finally, the concept shift skew includes two forms: the same label but different features and the same features but different labels.

In this study, we employ horizontal FL on non-IID data, which is based on the feature distribution skew characteristic.

2.3. FL on medical domain

The application of FL in the medical domain mostly deals with two tasks: classification and segmentation. The classification task is to clas-

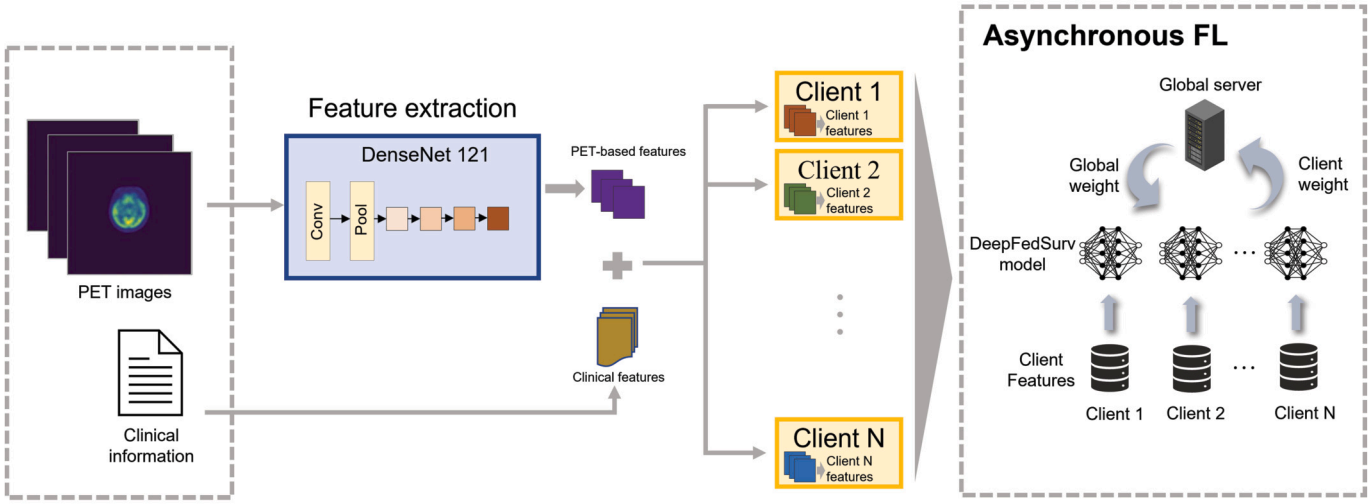


Fig. 1. Workflow of the proposed method on two separate datasets. First, we collected a public dataset from the Cancer Imaging Archive and an in-house dataset from Chonnam National University Hwasun Hospital. Then, we extracted features from PET images using DenseNet model. Next, we divided these extracted features with clinical features into various clients based on histological attributes and then integrated them into the FL framework.

sify a given set of instances with classes. Several classification studies have applied FL in medical field, such as cancer diagnosis [14], COVID-19 detection [40,51,43], patient hospitalization prediction [10], patient mortality prediction [39,44], and autism spectrum disorder (ASD) [37].

The segmentation task is the process of identifying a region of interest within a medical image. Medical input data can be in several forms, such as MRI [49,52], Ultrasound images [14,41], Tobacco and radon data [53], X-ray images [40,51,43], and CT [48]. There are several applications of FL studies in medical image segmentation such as brain tumor disease [49], COVID-19 region [48], and prostate region [52].

Our study is the first application of FL on multimodal (PET images and clinical risk factors) on predicting survival time of NSCLC patients. We believe that this study contributes to future clinical trials as it optimizes FL-based techniques for survival time prediction inputted with PET images and clinical risk factors.

3. Methods

In this paper, we utilized three training methods including single-client-based learning (CBL), centralized learning (CL) and FL. In the CBL framework, each client trains its own DL model using its local training dataset without data or model sharing from other clients. Thus, this CBL framework faces the data generalization problem when it is difficult to adapt properly to unseen data. Next, another traditional pipeline is the CL framework. In CL framework, all datasets from different sources are stored on a server and trained to build a global model. Therefore, it is sensitive and cannot ensure the privacy. In contrast to CBL and CL, FL does not share training data among clients but allows their collaboration by sharing their trained models to improve the global network performance. This enhances data privacy and security.

For easy visualization, the study design is shown in Fig. 1. First, a public dataset from the Cancer Imaging Archive (TCIA) and an in-house dataset from Chonnam National University Hwasun Hospital (CNUHH) were collected. Then, the features from PET images were extracted using the DenseNet model, which achieved good performance on our recent computer vision tasks for survival prediction [21]. Next, the combined features, including clinical features and PET-based features were divided into various clients based on histological features, and then a DL model was applied for predicting the survival time. Finally, we evaluated our framework in three different situations: CBL, CL and FL.

3.1. Dataset description

We examined clinical information and PET-based features to evaluate the proposed FL framework's ability to predict the survival time of NSCLC patients. In particular, we used a public radiogenic dataset of NSCLC (RNSCLC) from TCIA in addition to an in-house dataset from the CNUHH in South Korea. The patient demographics and clinical risk factors of the two NSCLC datasets are presented in Table 1. Besides that, Fig. 2 overviews the survival curve by stage on RNSCLC and CNUHH datasets.

The RNSCLC dataset consisted of clinical information and PET images for 211 patients. The clinical information consisted of seven features: age, sex, histology, pathology T stage, pathology N stage, pathology M stage, and smoking status. However, some clinical features were missing, and several patients lacked PET images. Therefore, a total of 34,207 PET images were used in our experiments.

To increase the generality of the method, we conducted another experiment on a larger dataset. The CNUHH dataset consisted of clinical information and PET images for 2,687 patients. The clinical information consisted of nine features: age, sex, histology, clinical T stage, clinical N stage, clinical M stage, overall stage, smoking history and smoking amount. In addition, the number of PET images varies between patients (approximately 300 images for each patient), for a total of 557,538 PET images.

3.2. Data preprocessing

Next, all data were pre-processed before further experiments. Given the raw clinical information from the two datasets, we first removed all irrelevant or duplicate data items. Notably, missing data refers to values that are not available for analysis and is a common issue encountered in many study designs. Examples of missing data include patients who were lost to follow-up or partially filled out forms [54]. It has been shown that high rates of missing data may negatively impact conclusions in primary care studies. Therefore, we removed all missing values from both clinical datasets. Furthermore, we converted all categorical data (such as sex, histology) into numerical data through techniques such as one-hot encoding, and normalized or standardized numerical data to ensure that it had a consistent scale.

DICOM-type PET images were converted into PNG-type images with three channels using the Pydicom library. Then, these images were normalized in the range [0,1] before being input to the DenseNet model for the feature extraction step.

Table 1
Patient demographic and clinical risk factors of two NSCLC datasets.

Variables	RNSCLC			Variables	CNUHH		
N		211	100%	N		2687	100%
Age (years)		67.96 ± 9.98		Age (years)		67.95 ± 9.63	
Sex				Sex			
	Male	135	63.98%		Male	2005	74.62%
	Female	76	36.02%		Female	682	25.38%
Histology				Histology			
	Squamous cell carcinoma	35	16.59%		Squamous cell carcinoma	1038	38.63%
	Adenocarcinoma	172	81.52%		Adenocarcinoma	1501	55.86%
	Large cell carcinoma	0			Large cell carcinoma	39	1.45%
	Unspecified NSCLC	4	1.89%		Unspecified NSCLC	109	4.06%
Smoking status				Smoking status			
	Yes	163	77.25%		Yes	1861	69.26%
	No	48	22.75%		No	826	30.74%
Survival status				Survival status			
	Censored	148	70.14%		Censored	830	30.89%
	Death	63	29.86%		Death	1857	69.11%
Overall survival time (days)				Overall survival time (days)			
	Q1	283	-		Q1	193.5	-
	Q2	600	-		Q2	568	-
	Q3	1065.5	-		Q3	1239	-
	Q4	2356	-		Q4	3805	-
	IQR	782.5	-		IQR	1045.5	-
	None	148	-		-	-	-
Pathological T stage				Clinical T stage			
	1	71	33.65%		0	0	0%
	2	57	27.01%		1	575	21.40%
	3	21	9.95%		2	1066	39.67%
	4	7	3.32%		3	544	20.25%
	Not collected	55	26.07%		4	502	18.68%
Pathological N stage				Clinical N stage			
	0	129	61.14%		0	930	34.61%
	1	15	7.11%		1	329	12.24%
	2	18	8.53%		2	709	26.39%
	Not collected	49	23.22%		3	719	26.76%
Pathological M stage				Clinical M stage			
	0	157	74.41%		0	1700	63.27%
	1	5	2.37%		1	987	36.73%
	Not collected	49	23.22%		-	-	-

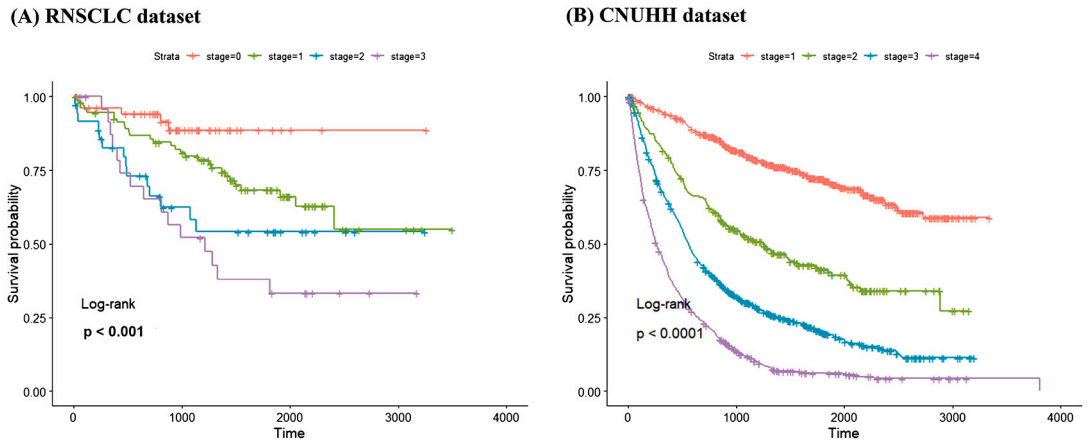


Fig. 2. Survival curves according to overall stage on (A) RNSCLC and (B) CNUHH datasets.

3.3. PET-based feature extraction

We employed DenseNet for PET-based features extraction given its successful application in our previous work [21,22]. In detail, the entire PET images of whole body (from the skull base to the thigh) were used as input and the corresponding survival time of each patient was used as a label for regression tasks. Then, the output layers of DenseNet were modified to obtain the features of PET images. The correlation of each PET-based feature with target survival time was evaluated by using mean absolute error (MAE) loss function as shown in Fig. 3. Because of the diversity in the number of PET images for each patient, the final variable features vector for each patient is calculated according to Equation (1).

$$P_i = \frac{1}{M}\Sigma_{t=1}^{M} p_t^{1\ldots n} \quad (1)$$

where P_i is the PET-based features of patient i_{th}, M is the number of PET images for each patient, and n is the number of valuable features for each PET image.

The model was trained through 30 epochs using Adam optimization, MAE loss function with a learning rate of 1e-3.

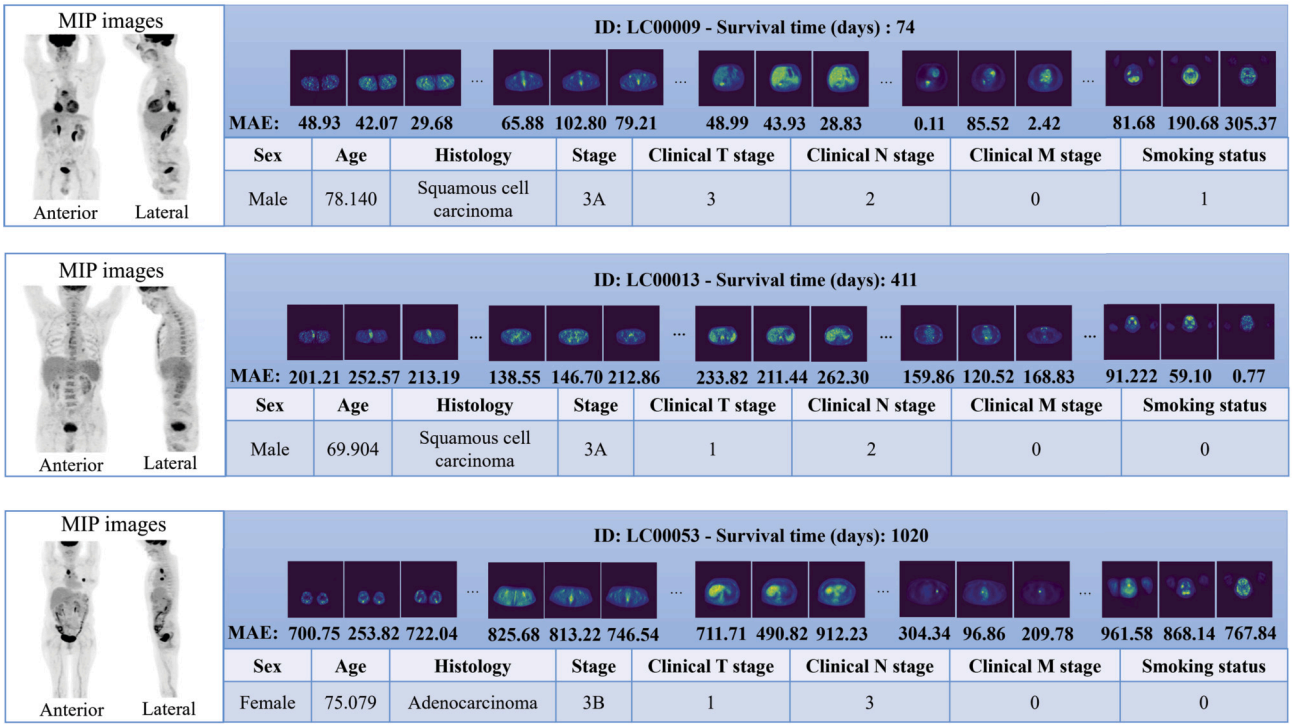


Fig. 3. Examples of valuable features including PET-based features and clinical risk factors. For PET-based features extraction, the entire PET images of whole body were used as input and the corresponding survival time of each patient was used as a label for regression task. The correlation of PET-based features and target survival time was performed by using MAE loss function. For clinical risk factors, all categorical data (such as sex, histology) were converted into numerical data through techniques such as one-hot encoding, and normalized or standardized numerical data to ensure that it had a consistent scale. PET data is visualized by using maximal intensity projection (MIP) method.

3.4. Deep neural network

We next developed a deep neural network named DeepFedSurv to perform predictive tasks on both PET-based and clinical features. Examples of valuables features including PET-based features and clinical risk factors are shown in Fig. 3. The DeepFedSurv architecture is presented in Fig. 4. The survival output denoted by Y is considered. A training set with labeled instances $T = \{(p_i, y_i)|p_i \in P \bigoplus (c_i, y_i)|c_i \in C, y_i \in Y\}$ is given, where $\bigoplus$ is the concatenation operator, and P and C are the PET-based and clinical input space, respectively. The target is to predict the survival time $y_i \in Y$ of an input $\{p_i \in P \bigoplus c_i \in C\}$. The value of node i_{th} in Dense layer k_{th} is calculated by Equation (2).

$$D_k^i = \max(0, \sum_{j=1}^{N}(D_{k-1}^j \times w_j + b_j)) \tag{2}$$

where N is the number of node, w is the weight vector mapping D_k to a real survival time and b is the bias.

Generally, the model was trained with an Adam optimizer [55] learning rate of 1e-3, and a MAE loss on each client.

3.5. FL

Fig. 5 shows the scheme of the asynchronous FL framework for survival time prediction. Each client (representing each medical institution) uses its own local data and trains with the DeepFedSurv model independently of other clients. After that, all client weights are uploaded to the global server for aggregation. Then, the global server transfers the weight update back to each client to achieve the learning goal. Accordingly, a collaborative global server and client are shown in Algorithm 1. This algorithm clearly presents the asynchronous operation in the FL framework where the training process on each client was performed independently and in parallel with other clients without having to wait for the training process on other clients being completed.

Algorithm 1 Federated Learning for individual survival time prediction.

- Input: PET-based vs clinical features
- Output: Global model weight

```
Initialize the global model weight W_G
for each round t = 1, 2, ..., R do
    for each client i = 1, 2, ..., N do
        Get global weight: W_L^i ← W_G
        Fit client model
        Update local weight W_L^i
        Scale model weight W_L^i = θ W_L^i
    end for
    Update the global weight using the aggregated function W_G^t = (1/N) Σ_{i=1}^{N} W_L^i
end for
return W_G
```

The weight of local models has a significant impact on FL performance [56,42,57]. Therefore, the scaling factor θ was designed to assign the appropriate weight to the new local model, where the weight of the local model increases with the value of θ. If the local model has higher accuracy than the global model, then θ will be increased to a higher weight and assigned to the local model. Therefore, we screened the scaling factor θ from 0 to 1 to achieve the best global model.

4. Experiments and results

4.1. Evaluation

The proposed method was evaluated using the MAE and concordance index (C-Index). Notably, the MAE is the measure of errors between the actual survival times and the predicted survival times in uncensored data, which is shown in Equation (3).

$$MAE = \frac{1}{n}\Sigma_{i=1}^{n}|\hat{y}_i - y_i| \tag{3}$$

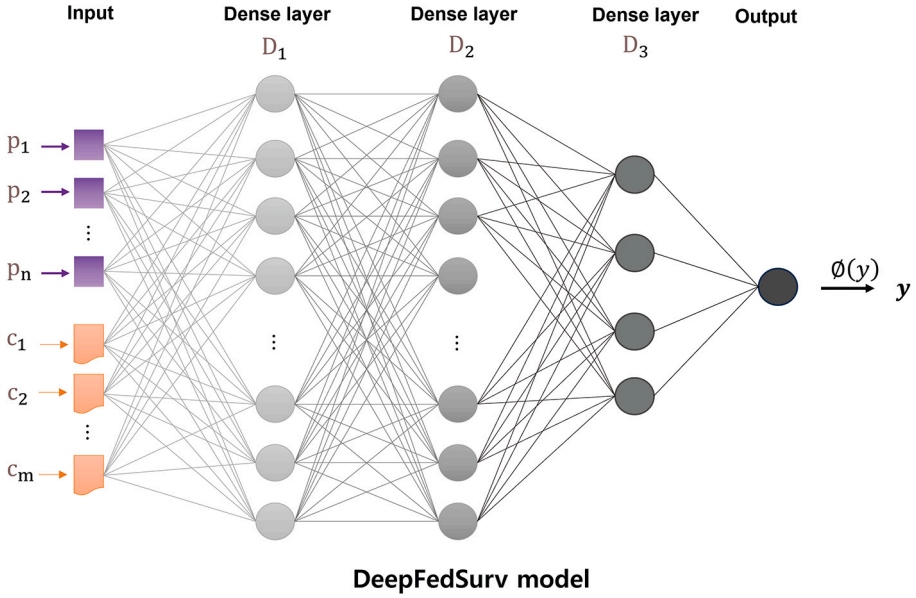


Fig. 4. The DeepFedSurv architecture. p_{ith} is the PET-based input th, c_{th} is the clinical input th and Ø is the Linear function.

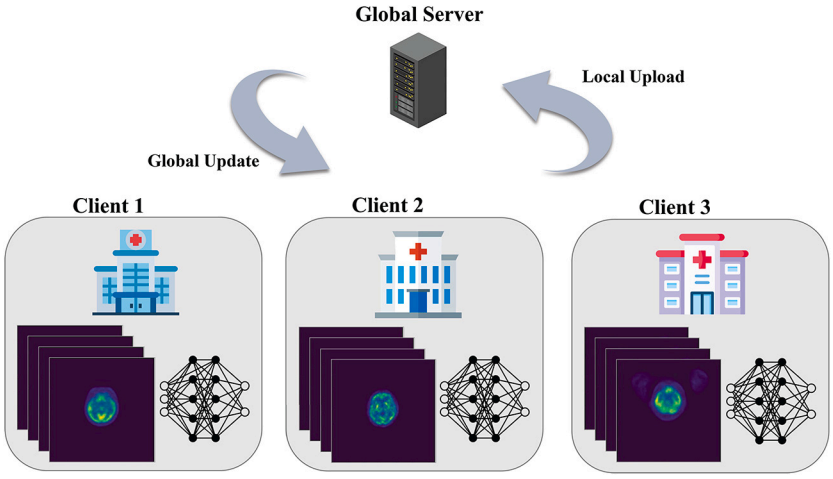


Fig. 5. Scheme of asynchronous FL framework for survival time prediction. The framework includes the global server and client server. Both servers use the same model architecture as FedSurv. Each client site trains using the FedSurv model on local data in parallel and sends the model weights for each round to the global server. On the global server, the global model is updated by averaging the model weights retrieved from all client sites. Then, the updated weights are sent back to each client model. The original client data are stored locally and cannot be exchanged or migrated.

where $\hat{y}$ is the predicted and y is the actual value. In addition, the C-Index is the rate of concordant pairs among all pairs of subjects, which is shown in Equation (4).

$$C - Index = \frac{\Sigma_{i,j}.I_{T_j<T_i}.I_{N_j>N_i}.\delta}{\Sigma_{i,j}I_{T_j<T_i}.\delta} \tag{4}$$

where N_i is the risk score of a unit i

$$1_{T_j<T_i} = 1 \text{ if } T_j < T_i \text{ else } 0$$

$$1_{N_j>N_i} = 1 \text{ if } N_j > N_i \text{ else } 0$$

4.2. Experiment setting

The two datasets were observed to investigate the role of FL in individual survival time prediction. Overall, there were a total of 2687 patients in the CNUHH dataset and 211 patients in the RNSCLC dataset. Each dataset was randomly divided into two sets: 80% for the training set and 20% for the global testing set (Tables 2 and 3). We conducted three main experiments: CBL, CL and FL.

For the CBL experiment, each client uses its local training dataset to train on it own DeepFedSurv model described in detail in Section 3.4. Then each client model was tested on global testing set as previously described.

For the CL experiment, we used the DeepFedSurv model to train on the whole training set and test on the global testing set.

For the FL experiment, we implemented horizontal FL algorithms based on non-IID characteristic. We found that the distribution of histology features varied from each dataset, as shown in Table 1. Therefore, we performed multiple sites using histology features. We first identified the histological source of each patient case in both datasets. We then partitioned the set of unique histological source sites into multiple nonoverlapping groups and grouped the data corresponding to each subset of histological source sites to serve as multiple distinct clients. The CNUHH training set was divided into four clients and the RNSCLC dataset was divided into three clients based on the histological characteristics of the patients. Notably, the global testing set included all

Table 2
RNSCLC dataset distribution.

Scenarios		Training set			Global test set
		Squamous cell carcinoma	Adenocarcinoma	Unspecified NSCLC	
CL		28	138	3	42
FL	Client 1	28	0	3	
	Client 2	0	138	0	
	Client 3	0	0	3	

Table 3
CNUHH dataset distribution.

Scenarios		Training set				Global test set
		Squamous cell carcinoma	Adenocarcinoma	Large cell carcinoma	Unspecified NSCLC	
CL		830	1201	31	87	538
FL	Client 1	830	0	0	0	
	Client 2	0	1201	0	0	
	Client 3	0	0	31	0	
	Client 4	0	0	0	87	

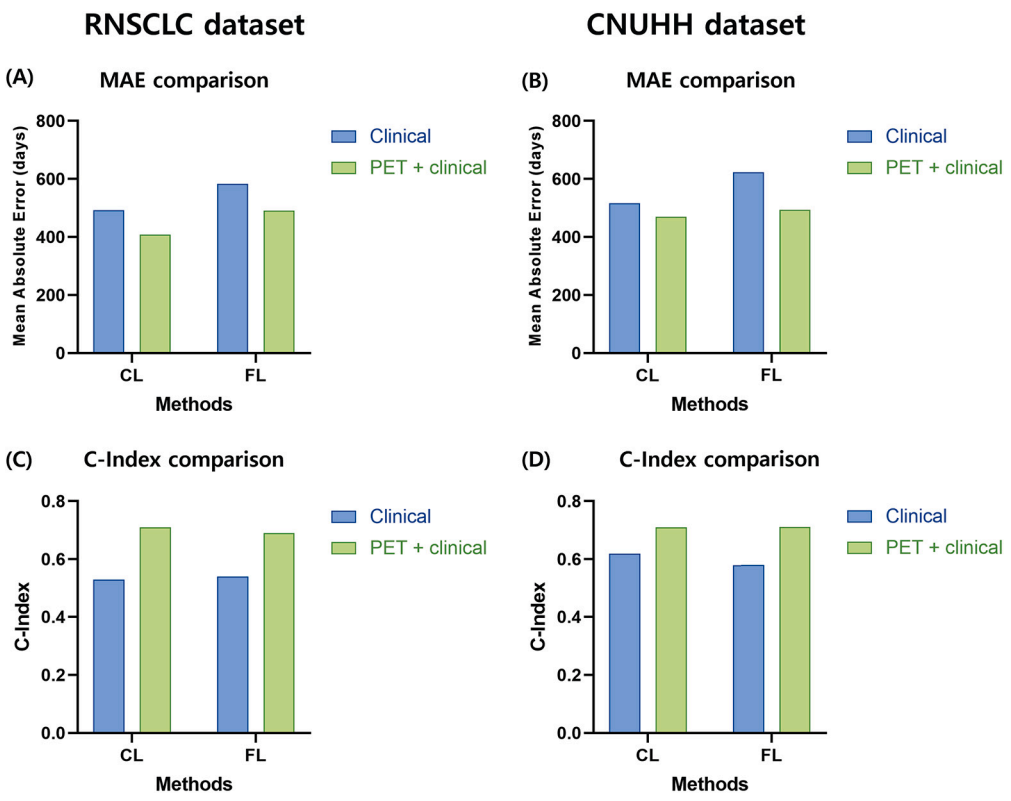


Fig. 6. Results comparison of PET-based and clinical features according to centralized learning (CL) and federated learning (FL) methods. (A) Performance in terms of MAE on CNUHH dataset. (B) Performance in terms of MAE on RNSCLC dataset. (C) Performance in terms of C-Index on the CNUHH dataset. (D) Performance in terms of C-Index on the RNSCLC dataset.

histological types. Once the client sites were identified, each data site was then randomly partitioned into a training set and a test set, respectively, consisting of 80% and 20% of the patient cases from each site. To train each client model, we used the DeepFedSurv model architecture with hyper-parameters as described in Section 3.4.

4.3. PET-based features for improving performance

We next demonstrated the effectiveness of the combination of PET-based and clinical features for predicting survival time, and the results are shown in Fig. 6. Overall, these results also demonstrated that the combination of clinical features and PET-based features helped to improve the individual survival time prediction performance compared with using only clinical features. Indeed, in the absence of PET-based features, the MAE increased nearly 20%, while the C-Index decreased around 30%. According to the experiment, PET-based features appeared to lead to a performance increase. Collectively these data confirmed that PET-based features play a noteworthy role in individual survival time prediction.

4.4. CBL, CL and FL comparison

FL is typically designed for decentralized data that cannot be easily accessed in a centralized location. However, it is often convenient to conduct initial experiments using data that can be downloaded and manipulated locally at the research and development stages. In our study,

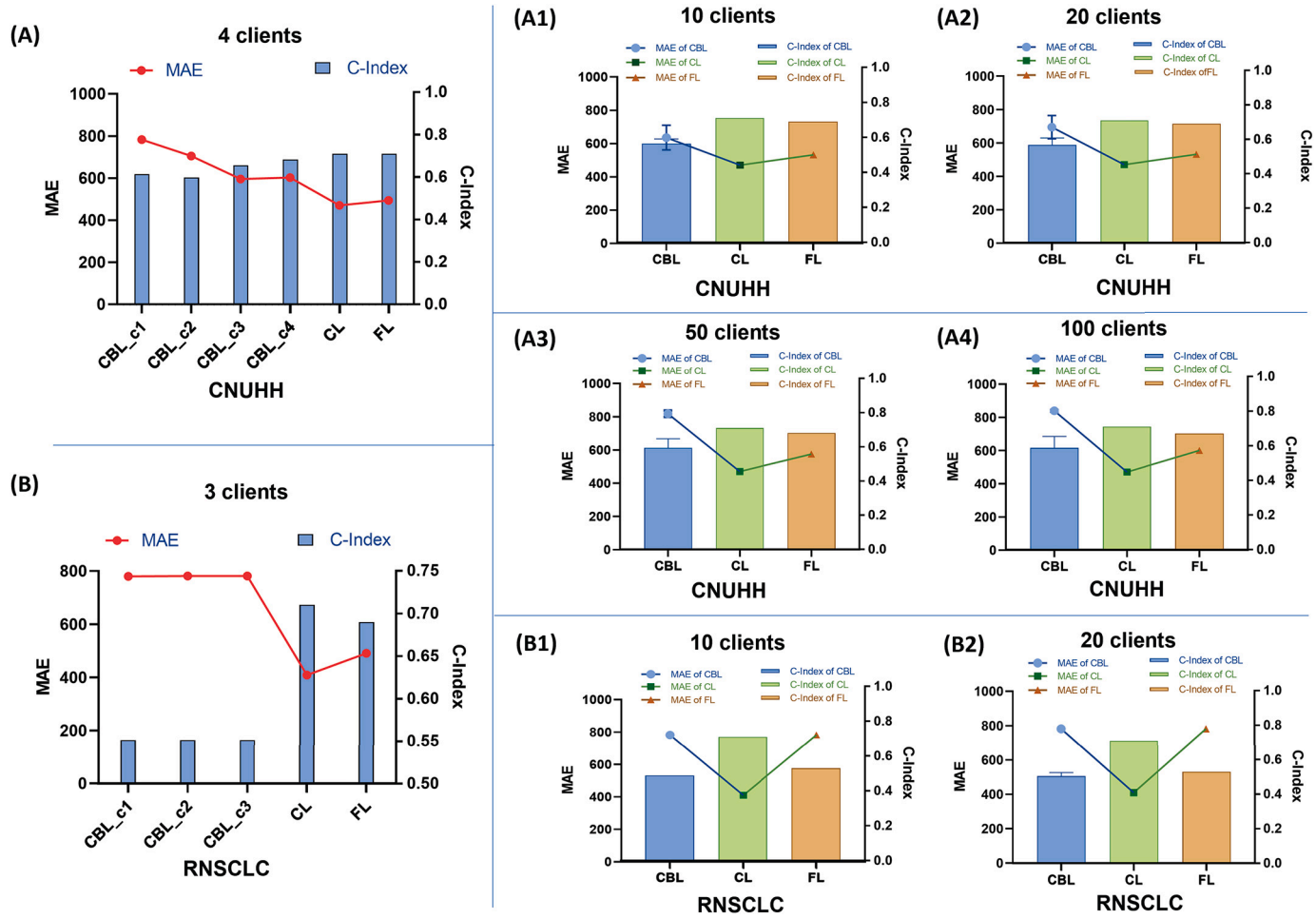


Fig. 7. Results comparison of three training approaches: CBL (client-based learning), CL (centralize learning), and FL (federated learning) on RNSCLC and CNUHH global test sets. Lower MAE and higher C-Index values indicate better performance.

we compared the individual survival time prediction performance on the RNSCLC dataset and the CNUHH dataset using CL and FL. The results in Tables 4 and 5 show that our FL-based method achieved competitive performance in a centralized setting using PET-based and clinical features (Federated: 490.80 days, Centralized: 409.10 days in Table 4; Federated: 494.25 days, Centralized: 470.60 days in Table 5). Although the performance with FL was not higher than that with CL, the difference between the two was not substantial, indicating that FL is still effective in individual survival time prediction.

Fig. 7 depicts the comparison of different approaches (CBL, CL and FL) on RNSCLC and CNUHH global test sets in terms of two metrics. The CBL approach had higher MAE and a lower C-Index results than the other approaches in all cases. Besides that, the CL approach achieved the best performance in term of two metrics. Furthermore, on CNUHH dataset, the FL obtained the competitive results to CL approach with various numbers of clients (Fig. 7.A, A1, A2, A3, A4). With three clients on RNSCLC dataset (Fig. 7.B), the FL outperformed all CBL approaches and achieves competitive results with CL. However, with 10 and 20 clients (Fig. 7.B1, B2), the results were not improved much when using FL approach. This can be explained by the lack of data in RNSCLC dataset.

4.5. Ablation studies

Fig. 8 shows the comparison of MAE results on PET-based and clinical features by scale according to four different training strategies with different numbers of epochs per round. The global model converged after 10 rounds. MAE values tended to decrease as the scale values increased. In particular, the highest scaling value ($\alpha = 0.25$) showed the lowest MAE value in all epoch cases, and the lowest scaling value ($\alpha = 0.05$) showed the highest MAE value in all epochs. We thus concluded that to achieve the best MAE value, the model weight must be suitable (in this case, scaling factor $\alpha = 0.25$ for training on clinical and PET features).

In addition, we performed an extensive ablation study in four settings to validate the performance and robustness of the proposed method. In particular, we randomly decomposed each dataset into 10, 20, 50 and 100 clients instead of decomposing the training set into three or four clients based on histological attributes. Again, we used the DeepFedSurv model presented in Section 3.4 and the FedSurv framework. The ablation results of our method are shown in Table 6. Due to the lack of data in the RNSCLC dataset, only 10 and 20 clients were explored. In the CNUHH dataset, the best result was achieved with 10 clients (507.12 MAE and 0.69 C-Index) and the worst result was achieved with 100 clients (600.89 MAE and 0.67 C-Index). It can be seen that there was little variance in terms of both MAE and C-Index when the number of clients differed. In other words, our proposed model worked stably regardless of the number of clients.

5. Discussion

Survival analysis is an important step in planning optimal cancer treatment strategies. In our previous studies [21,22], we focused on survival prediction for patients with NSCLC using CNN (e.g., DenseNet, ResNet) combined with CoxPH [25]. These models were employed to predict individual survival distribution and time using clinical risk fac-

Table 4
Individual survival time prediction performance of the RNSCLC dataset, reported as MAE and C-Index ($\pm sd$). α is the scaling factor in the FL setting. Lower MAE and higher C-Index values indicate better performance.

Method	Clinical		PET + Clinical	
	MAE ↓	C-Index ↑	MAE ↓	C-Index ↑
Centralized	492.95 ± 136.17	0.53 ± 0.01	409.10 ± 118.36	0.71 ± 0.02
Federated ($\alpha = 0.05$)	781.41 ± 0.01	0.02 ± 0.01	780.37 ± 0.05	0.68 ± 0.01
Federated ($\alpha = 0.10$)	777.56 ± 0.02	0.54 ± 0.01	774.28 ± 0.06	0.69 ± 0.01
Federated ($\alpha = 0.15$)	781.56 ± 0.02	0.50 ± 0.01	738.68 ± 1.36	0.68 ± 001
Federated ($\alpha = 0.20$)	750.30 ± 3.77	0.54 ± 0.01	703.46 ± 7.59	0.50 ± 0.01
Federated ($\alpha = 0.25$)	583.83 ± 30.01	0.54 ± 0.01	490.80 ± 22.95	0.69 ± 0.01

Table 5
Individual survival time prediction performance of the CNUHH dataset, reported as MAE and C-Index ($\pm sd$). α is the scaling factor in the FL setting. Lower MAE and higher C-Index values indicate better performance.

Method	Clinical		PET + Clinical	
	MAE ↓	C-Index ↑	MAE ↓	C-Index ↑
Centralized	516.11 ± 21.81	0.62 ± 0.06	470.60 ± 4.80	0.71 ± 0.00
Federated ($\alpha = 0.05$)	837.22 ± 0.01	0.50 ± 0.01	834.06 ± 0.06	0.70 ± 0.01
Federated ($\alpha = 0.10$)	801.69 ± 0.67	0.53 ± 0.01	776.45 ± 0.44	0.50 ± 0.00
Federated ($\alpha = 0.15$)	707.66 ± 3.48	0.55 ± 0.01	663.94 ± 1.31	0.71 ± 0.01
Federated ($\alpha = 0.20$)	613.02 ± 6.29	0.57 ± 0.01	540.54 ± 6.25	0.71 ± 0.01
Federated ($\alpha = 0.25$)	623.54 ± 35.55	058 ± 0.01	494.25 ± 40.16	0.71 ± 0.01

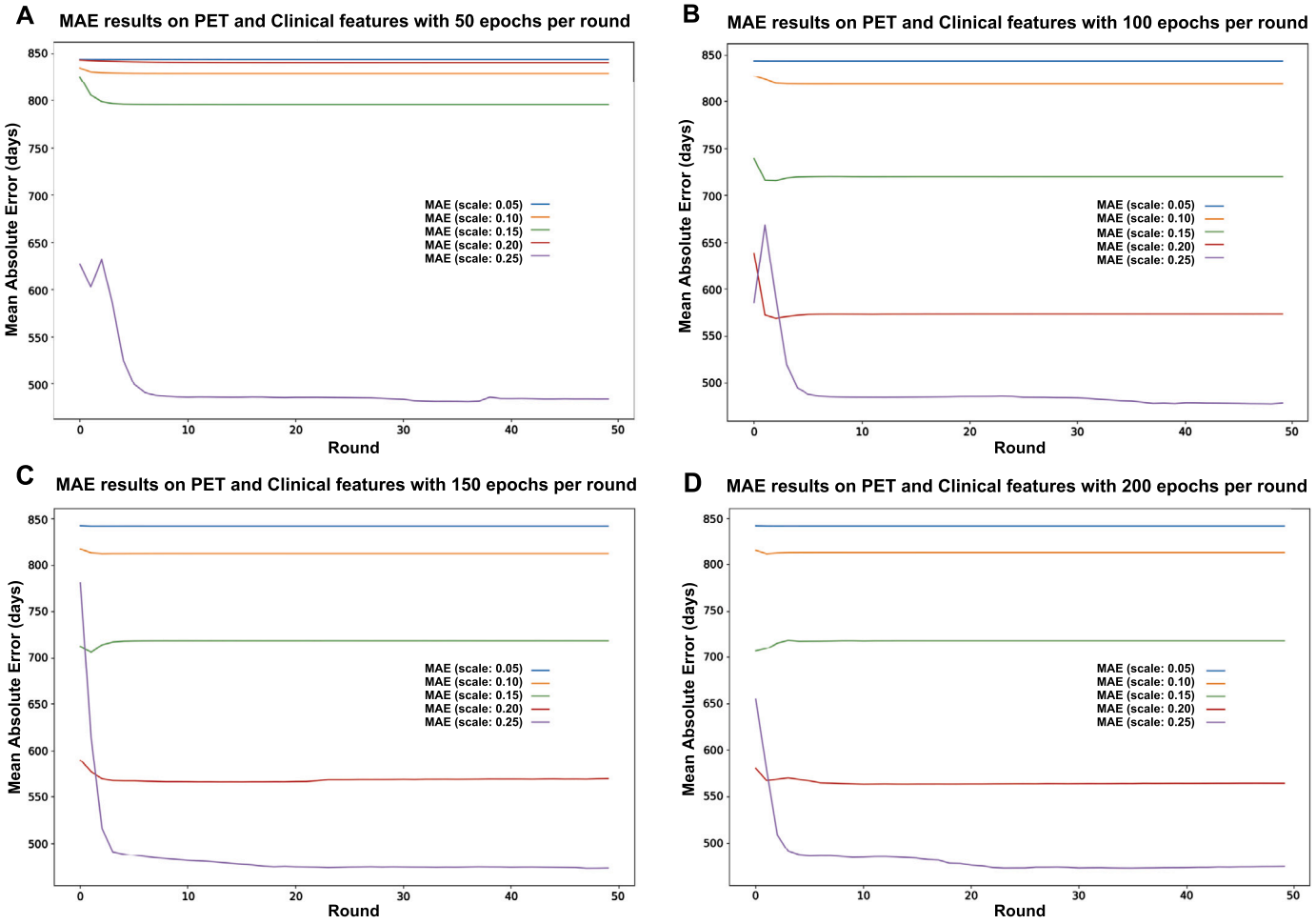


Fig. 8. MAE results comparison on PET-based and clinical features by scale according to four different training strategies with different numbers of epochs per round on CNUHH dataset. (A) MAE results on PET-based and clinical features with 50 epochs per round. (B) MAE results on PET-based and clinical features with 100 epochs per round. (C) MAE results on PET-based and clinical features with 150 epochs per round. (D) MAE results on PET-based and clinical features with 200 epochs per round.

Table 6
Ablation results of FedSurv framework on RNSCLC and CNUHH datasets for 10, 20, 50 and 100 clients.

#Clients	RNSCLC		CNUHH	
	MAE ↓	C-Index ↑	MAE ↓	C-Index ↑
10 clients	781.77 ± 0.01	0.53 ± 0.01	507.12 ± 60.28	0.69 ± 0.02
20 clients	781.77 ± 0.01	0.53 ± 0.01	531.79 ± 60.94	0.69 ± 0.03
50 clients	-	-	575.75 ± 30.76	0.68 ± 0.03
100 clients	-	-	600.89 ± 25.08	0.67 ± 0.01

tors and FDG PET images, which can be crucial to predict prognosis given their inherent information on the metabolic activity of cancer cells. In addition, FDG PET is a powerful tool to investigate the role of tumoral metabolic activity in whole body [58], specially predicting prognosis of lung cancer [59]. Therefore, we investigated the effectiveness of the entire PET images of whole body to the survival time prediction of lung cancer patients. In this study, we developed an FL-based framework able to predict individual survival time depending on both FDG PET-based and clinical features using two independent NSCLC datasets.

Previous studies commonly built AI-based survival models by outputting the survival probability or cumulative risk [21,22,60–62]. It has been shown that these methods, which use censored and uncensored data, are evaluated by using common evaluation metrics such as CI, Brier score, IBS (integrated Brier score), IAE (integrated absolute error) and ISE (integrated square error). MAE is rarely used in these practices. It is important to note that our approach, which uses only uncensored data, requires the model to output survival time. Therefore, we employed MAE and CI measurements to evaluate our method, which can automatically predict the survival time from PET-based and clinical data in NSCLC patients.

The performance of survival prediction can be explained by the DL-based algorithm in a single process [21,22,60–62]. We further investigated how to integrate DL and FL. We then examined results from three scenarios where we either trained all samples using only the DL model as CBL and CL, or trained from multiple clients using DL and FL. However, choosing the right data partition and distribution remains a challenge for FL during the research phase. In this study, we implemented non-IID data between different clients, which is horizontal FL algorithms. Besides that, we applied the scaling factor to the FL framework to address heterogeneity in various clients. In our study, the FL approach achieved centralized method performance in PET-based survival time prediction and outperformed single-client-based approaches. The proposed framework has the potential to be applied to various scenarios that involve private user data, such as medical information.

Our approach provides several advantages compared to conventional methods. First, we demonstrated the feasibility and effectiveness of the FL approach for survival time prediction using large-scale multimodal datasets. In addition, by integrating FL with DL, we successfully addressed the potential of FL in survival time prediction for patients diagnosed with NSCLC. This effort was undertaken to address the increasing demand for large-scale and inter-institutional studies, contributing to advancements in medical research.

However, our work also has the following limitations. First, our framework only achieved competitive results on small and medium datasets and therefore needs to be explored with further experiments on larger datasets. Second, the data privacy of our framework needs to be improved since preserving the privacy of larger datasets usually requires more robust protection [63]. In future, we will explore the application of our framework to real-world hospitalization systems to analyze the survival of NSCLC patients in a way that both ensures efficiency and preserves patients' privacy.

6. Conclusions

In conclusion, we successfully demonstrated the feasibility and effectiveness of FL for the general-purpose prediction of survival time using clinical information and PET-based information. Additionally, we presented the effectiveness of combining PET-based and clinical features for the prediction of survival time in patients with NSCLC. To the best of our knowledge, this is the first study to utilize FL in the field of PET-based survival time prediction. We believe that our FL framework has the potential for application to crucial medical image analysis tasks beyond the scope of this study. These techniques may prove valuable in situations where transferring large numbers of medical images is restricted due to institutional or governmental regulations. The framework developed in this study will enable institutions to retain greater control over their data while still achieving performance levels that are on par with or close to those of state-of-the-art methods.

CRediT authorship contribution statement

Vi Thi-Tuong Vo: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Tae-ho Shin:** Writing – original draft, Visualization, Investigation. **Hyung-Jeong Yang:** Writing – review & editing, Writing – original draft, Validation, Methodology, Funding acquisition, Conceptualization. **Sae-Ryung Kang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Soo-Hyung Kim:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Additional tables

To gain a full understanding of three training approaches (CBL, CL, and FL) on RNSCLC and CNUHH global test sets, we report results in Tables A1, A2, A3, and A4.

Table A1
Ablation study results of three training approaches: CBL (client-based learning), CL (centralized learning), and FL (federated learning) on CNUHH dataset with four clients.

Methods	MAE	C-Index
CBL_c1	782.80	0.61
CBL_c2	704.76	0.60
CBL_c3	595.99	0.65
CBL_c4	602.32	0.68
CL	470.60	0.71
FL	494.25	0.71

Table A2
Ablation study results of three training approaches: CBL (client-based learning), CL (centralized learning), and FL (federated learning) on CNUHH dataset with 10, 20, 50 and 100 clients.

No. clients	10 clients		20 clients		50 clients		100 clients	
Metrics	MAE	C-Index	MAE	C-Index	MAE	C-Index	MAE	C-Index
Mean of CBL	635.64	0.56	694.57	0.57	818.20	0.59	839.44	0.59
CL	470.60	0.71	470.60	0.71	470.60	0.71	470.60	0.71
FL	531.79	0.69	531.79	0.69	575.75	0.68	600.89	0.67

Table A3
Ablation study results of three training approaches: CBL (client-based learning), CL (centralized learning), and FL (federated learning) on RNSCLC dataset with three clients.

Methods	MAE	C-Index
CBL_c1	780.21	0.55
CBL_c2	781.37	0.55
CBL_c3	781.37	0.55
CL	409.10	0.71
FL	490.80	0.69

Table A4
Ablation study results of three training approaches: CBL (client-based learning), CL (centralized learning), and FL (federated learning) on RNSCLC dataset with 10 and 20 clients.

No. clients	10 clients		20 clients	
Metrics	MAE	C-Index	MAE	C-Index
Mean of CBL	781.68	0.49	781.72	0.50
CL	409.10	0.71	409.10	0.71
FL	781.77	0.53	781.77	0.53

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